Problem 3 - Backyard Toolshed Routine

Week 7 | Category B Assignment | Java solutions

Problem

Build GardenTool <- CuttingTool <- Pruner. Each concrete use() override calls the parent implementation first.

Complete Java Code

```
abstract class GardenTool {
    public abstract String use();
}

class CuttingTool extends GardenTool {
    public CuttingTool() {
        super();
    }

    @Override
    public String use() {
        return "Using the tool in the garden, blade sharpened first";
    }
}

class Pruner extends CuttingTool {
    public Pruner() {
        super();
    }

    @Override
    public String use() {
        return super.use() + ", then trimming branches precisely";
    }
}

public class Main {
    public static void main(String[] args) {
        CuttingTool c = new CuttingTool();
        Pruner p = new Pruner();
        System.out.println(c.use());
        System.out.println(p.use());
    }
}
```

Sample Output

```
Using the tool in the garden, blade sharpened first
Using the tool in the garden, blade sharpened first, then trimming branches precisely
```

How it works

CuttingTool provides the first concrete use() routine. Pruner calls super.use() and appends only its own trimming detail, so the chain of behavior is reused.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.